SynGS: Synergizing explicit geometry and transformer depth priors for sparse-view 3D Gaussian Splatting

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ABSTRACT

In 360-degree scenes captured with sparse views, 3D Gaussian Splatting (3DGS) reconstruction is susceptible to the loss of edge detail, geometric instability in occluded regions, and overfitting to foreground regions. To mitigate these challenges under sparse-view conditions, we present SynGS, a 3D Gaussian Splatting method guided by geometric and depth priors for sparse-view 3D reconstruction. The proposed method leverages the visual hull to impose explicit geometric constraints and integrates a Transformer-based depth prior to facilitate global cross-view consistency, complemented by a geometry enhancement pipeline. By combining reliable depth filtering with fine-grained structural detail recovery and incorporating a dynamic visibility regularization strategy, our approach enables adaptive reweighting of supervision signals during optimization. Empirical evaluations demonstrate that SynGS achieves competitive performance, particularly in PSNR and SSIM. On Mip-NeRF 360, SynGS demonstrates consistent improvements in PSNR and SSIM across the evaluated sparse-view settings, with gains of 3.56 dB in PSNR and 0.025 in SSIM over a leading baseline using only four input views. On LLFF, the improvements are more pronounced in the four- and six-view settings, while the nine-view results remain comparable to those of leading existing methods.

Keywords: Novel View Synthesis, 3D Gaussian Splatting, Few-shot reconstruction, Geometry Prior-Guided Reconstruction, Vision Transformer

1. INTRODUCTION

Sparse inputs provide limited information about visible regions, causing 3D reconstruction methods to overfit local observations and struggle to maintain global geometric consistency.[1] Sparse observations also introduce greater geometric ambiguity, making occluded regions, textureless areas, and view boundaries prone to missing geometry, structural distortions, and degraded appearance.[2] Kerbl et al.[3] introduced 3D Gaussian Splatting (3DGS); with tiled rasterization it enables rapid training and real-time rendering, and subsequent variants improve structured, alias-free rendering.[4,5] Under sparse views, Structure-from-Motion (SfM)[6] often recovers incomplete point clouds. Since 3DGS relies on SfM cameras and sparse points for initialization, weak coverage harms geometric convergence.[7] Limited cross-view constraints further cause floaters and pseudo-geometry,[8] while sparse supervision[9] and visibility imbalance aggravate overfitting.[10]

Existing sparse-view methods have explored depth regularization,[11] Gaussian densification,[12] co-regularization,[13] and Gaussian dropout[9] to improve reconstruction. Nevertheless, incomplete initial geometry and overfitting during optimization remain challenges when only a few observations are available. To address these issues, we propose SynGS, a sparse-view 3DGS framework that combines geometry-guided Gaussian initialization with training-time regularization. By strengthening initial geometric support and mitigating overfitting during optimization, SynGS improves reconstruction robustness under severely limited view coverage. The main contributions are:

- **Visual Hull Initialization.** We incorporate a visual hull derived from multi-view silhouettes to provide an explicit geometric constraint for Gaussian initialization. This constraint complements incomplete SfM observations and helps establish a more reliable initial spatial distribution of Gaussian primitives.

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- **VGGT-Guided Densification.** We develop a pre-optimization densification strategy that back-projects cross-view-consistent depth estimates from VGGT to supplement the sparse initial point set. By introducing additional geometric samples before training, this strategy improves geometric coverage in under-observed regions.

- **Dynamic Visibility Regularization.** We incorporate Dynamic Visibility Regularization during training to stochastically mask Gaussian primitives, redistribute optimization gradients, and mitigate overfitting under limited-view supervision. This regularization complements the geometry-enhanced initialization to improve the robustness of sparse-view reconstruction.

2. RELATED WORK

Neural Radiance Fields (NeRF) represent scenes as implicit radiance fields and achieve high-quality novel view synthesis through differentiable rendering.[14] Under sparse views, however, insufficient observations cause severe geometric ambiguity. Depth-guided approaches further improve structural consistency,[15,16] while other methods constrain unobserved regions,[2] jointly refine noisy poses,[17] or leverage diffusion priors.[18] Grid-based anti-aliased variants such as Zip-NeRF further improve efficiency and rendering fidelity,[19] and baked neural surfaces enable real-time mesh-based view synthesis.[20] 3D Gaussian Splatting (3DGS) introduces an explicit scene representation based on optimizable Gaussian primitives and enables efficient training and real-time rendering through differentiable rasterization.[3] Beyond the original formulation, structured and anti-aliased Gaussian representations improve robustness and multi-scale rendering,[4,5] while surface-aligned variants target more accurate geometry recovery.[21,22] Depth-based approaches introduce additional geometric supervision,[11] while densification-based methods improve Gaussian coverage.[12,23] Other strategies include progressive geometric propagation,[24] gradient-aware density control,[25] and epipolar depth priors.[26] Others explore alternating densification under sparse inputs,[27] score distillation,[28] co-regularization,[13] and Gaussian dropout.[9,29] Pose-free and feed-forward sparse Gaussian reconstruction has also been studied.[30–33]

Classical multi-view stereo remains a strong geometric backbone.[34] Monocular depth estimators provide useful cues for underconstrained views,[35] and recent monocular geometry models further recover metric structure from a single image.[36] Although depth supervision, Gaussian densification, and training-time regularization have improved sparse-view reconstruction, limited-view observations can still result in incomplete initial geometry and overfitting during optimization. To address these challenges, SynGS combines explicit visual-hull constraints with a VGGT-based multi-view depth prior[37] to augment the initial Gaussian geometry before optimization, and incorporates Dynamic Visibility Regularization during training to alleviate sparse-view overfitting.

3. METHOD

3.1 Overall Framework

We propose SynGS, a sparse-view 3D Gaussian Splatting[3] framework that integrates explicit geometry initialization, depth-guided densification, and visibility-aware optimization. As illustrated in Figure 1, the input consists of sparse-view images and corresponding foreground masks. While recent variants organize Gaussians with structured scaffolds or octrees,[4,38] SynGS focuses on silhouette-based initialization and depth-guided densification under sparse posed inputs.

First, **Visual Hull Initialization** constructs a coarse geometric prior from multi-view silhouettes and generates an initial point cloud. Then, **VGGT-Guided Densification** exploits cross-view depth priors to supplement missing geometry and produce a depth-enhanced point cloud for Gaussian initialization. Finally, **Dynamic Visibility Regularization (DVR)** is applied during 3DGS optimization to rebalance supervision under sparse visibility and reduce overfitting. The complete process proceeds from the Visual Hull point cloud, through depth densification and Gaussian initialization, to DVR-regularized optimization.

3.2 Visual Hull Initialization

Under sparse views, Structure-from-Motion (SfM)[6] often produces sparse and incomplete point clouds. Weak coverage then destabilizes 3DGS optimization. We construct a Visual Hull prior from sparse multi-view images, camera parameters, and foreground masks, with light mask cleanup at boundaries. With known cameras, silhouettes are back-projected into 3D, and multi-view silhouette consistency defines a coarse geometric envelope of the target.[39]

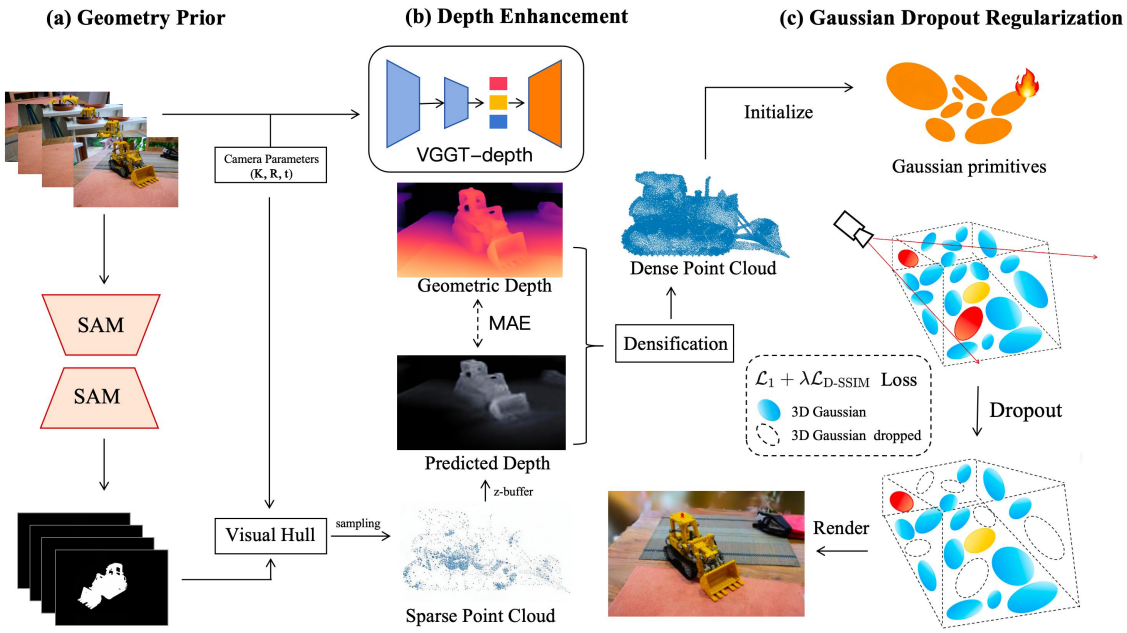


Figure 1. Overview of the SynGS pipeline. The third stage applies Dynamic Visibility Regularization (DVR).

Candidate points are retained only when their projections agree with all valid foreground masks, then cleaned by one-time statistical and radius-based outlier filtering. The scaffold is stable but coarse: silhouette-based reconstruction cannot fully recover concave regions and fine structures. Sec. 3.3 supplements the missing geometry before Gaussian initialization.

3.3 VGGT-Guided Densification

Silhouettes alone miss concave regions and fine structures. We densify the Visual Hull point cloud with **VGGT-Depth**: from a pre-trained VGGT backbone,[37] a depth prior adaptation branch with cross-attention over multi-view tokens and a lightweight fusion head predicts per-view depth and confidence. Unlike a frozen monocular depth network,[35] the adaptation branch is conditioned on multi-view tokens (Figure 2).

We use the Visual Hull AABB diagonal L as the scene-scale reference. For each view, masked depth MAE against dataset ground-truth depth is used only to filter views and rays during densification (not at inference or when computing metrics); it is normalized by the 95th percentile over validation views to obtain q_v. Views with $q_v < 0.4$ are discarded. If fewer than $0.8N$ views remain, densification is skipped and the Visual Hull point cloud is retained. Remaining views keep regions that pass VGGT-Depth confidence and depth consistency. With tolerance

$$\Delta(d) = \alpha d + \beta, \qquad \alpha = 0.01, \quad \beta = 0.002L, \tag{1}$$

we reject low-confidence or inconsistent observations before back-projection.

Point supplementation runs only along reliable rays. For each point, the top $K = 3$ reliable views are selected; candidates are sampled around the predicted depth and back-projected. Related densification strategies propagate geometric cues or control Gaussian density during optimization;[24,25] alternating densification schedules have also been studied for sparse inputs.[27] Here supplementation is applied offline to the Visual Hull point cloud before training. Complementary work injects epipolar depth priors into sparse-view Gaussian reconstruction,[26] or rasterizes depth from Gaussians for geometric supervision;[40] we instead use adapted VGGT-Depth predictions for back-projection. Supplementary points are merged via two-stage voxel fusion and deduplication ($0.01L$ then $0.004L$).

Each densified point initializes a Gaussian center; scale comes from the mean distance to its $k = 8$ nearest neighbors. Multi-view colors are fused with depth-consistency and VGGT confidence weights. Opacity is set from cross-view consistency and clipped to $[0.2, 0.8]$. Uneven supervision remains under sparse views; we address this with DVR (Sec. 3.4).

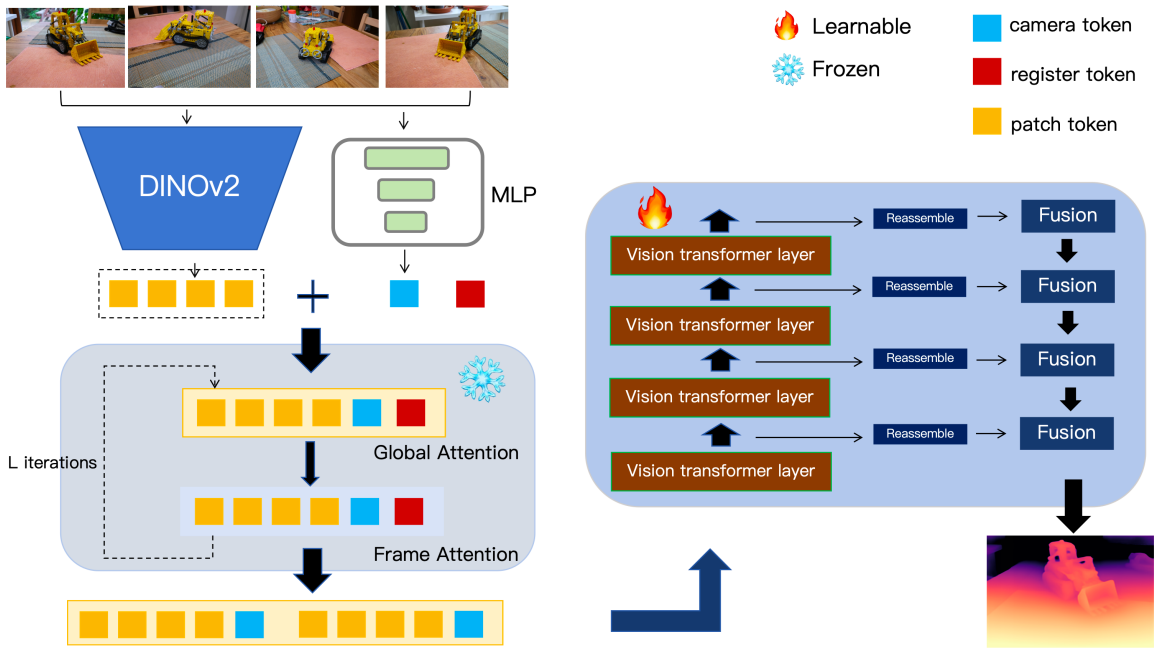


Figure 2. VGGT-Depth prior used for depth-guided point cloud densification.

3.4 Dynamic Visibility Regularization

Sparse-view optimization remains strongly visibility-dependent. Distant, occluded, or poorly observed Gaussians appear in fewer views or pixels and receive weaker reconstruction gradients.[8,9] Visibility priors have also been explored to regularize sparse-input radiance fields,[10] and distractor-aware training further highlights sensitivity to unreliable observations.[41] Beyond fixed densification heuristics, densification can also be cast as stochastic sampling,[42] while constraining the Gaussian budget motivates careful control of which primitives participate in rendering.[43] We use DVR via progressive Gaussian dropout.

At iteration t, DVR randomly masks a subset of Gaussians for dropout rendering. The dropout ratio grows linearly:

$$r_t = \min\left(r_{\max}, \gamma \frac{t}{T}\right), \tag{2}$$

where t and T are the current and total iterations, $r_{\max}$ is the maximum dropout ratio, and $\gamma = 0.1$–0.2 controls the growth rate. Early training keeps weak dropout for geometric convergence; later training increases it against sparse-view overfitting.

Used alone, random dropout can induce opacity or scale compensation; we keep a full-render branch as an anchor. Unlike compensated rendering that rescales retained primitives,[9] the dropout branch is left uncompensated, while the full set is rendered in parallel under the same target. Let I_{full} and I_{drop} denote the full and dropout renderings, and I the ground-truth image. We use the standard color loss

$$\mathscr{L}_{\text{color}}(\hat{I}, I) = (1-\lambda)\,\mathscr{L}_1(\hat{I}, I) + \lambda\,\mathscr{L}_{\text{D-SSIM}}(\hat{I}, I), \tag{3}$$

with $\lambda = 0.2$.[3] The final objective is

$$\mathscr{L} = \mathscr{L}_{\text{color}}(I_{\text{full}}, I) + \beta\,\mathscr{L}_{\text{color}}(I_{\text{drop}}, I), \tag{4}$$

where $\beta = 0.5$–1.0. Related sparse-view designs further modulate masking with depth and density cues,[44] or drop anchors and spherical harmonics;[29] DVR instead uses a progressive random schedule with an uncompensated dropout branch.

4. EXPERIMENTS

4.1 Experimental Setup and Datasets

We evaluate SynGS on Mip-NeRF 360 (unbounded 360° scenes)[45] and LLFF (forward-facing real captures).[46] All images are downsampled by a factor of 8. Mip-NeRF 360 scenes are tested with 4, 6, and 9 uniformly sampled input views. Metrics are PSNR, SSIM, and LPIPS. We compare against the baselines in Tables 1 and 2. Table 1 includes RegNeRF,[2] SparseNeRF,[15] 3DGS,[3] CoR-GS,[13] DropGaussian,[9] and D^{2}GS.[44] Table 2 uses the same 3DGS-based baselines, with FreeNeRF[1] in place of SparseNeRF among the NeRF-based methods.

Training follows the standard 3DGS framework. Depth priors come from VGGT-Depth. Consistent with Sec. 3.3, view- and ray-level selection during densification may use dataset ground-truth depth only as a filter; novel-view rendering and all reported metrics do not use ground-truth depth. The model is trained for 30K iterations, with 3DGS initialization optimized for 10K iterations. Floaters are periodically removed during training. Experiments use PyTorch on an NVIDIA RTX 4090 GPU.

4.2 Quantitative and Qualitative Results

Table 1. Performance comparisons of sparse-view synthesis on Mip-NeRF 360 (4/6/9 views). LPIPS* denotes LPIPS $\times 10^2$.

	Methods	Mip-NeRF 360 (4-view)			Mip-NeRF 360 (6-view)			Mip-NeRF 360 (9-view)		
		LPIPS*↓	PSNR↑	SSIM↑	LPIPS*↓	PSNR↑	SSIM↑	LPIPS*↓	PSNR↑	SSIM↑
NeRF-based	RegNeRF	19.88	12.59	0.841	20.72	13.41	0.847	19.70	13.68	0.852
	SparseNeRF	17.76	12.83	0.845	19.74	13.42	0.861	21.56	14.36	0.873
3DGS-based	3DGS	10.80	20.31	0.899	8.38	22.12	0.913	6.42	24.29	0.930
	CoR-GS	11.76	20.45	0.856	9.28	22.57	0.901	7.04	24.73	0.926
	DropGaussian	11.57	21.76	0.837	8.61	22.98	0.895	8.02	24.10	0.915
	D^{2}GS	2.58	22.35	0.923	2.34	24.74	0.937	2.11	27.13	0.941
	SynGS (Ours)	3.89	25.91	0.948	3.43	27.93	0.953	3.02	29.21	0.961

Table 2. Performance comparisons of sparse-view synthesis on LLFF (4/6/9 views). LPIPS* denotes LPIPS $\times 10^2$.

	Methods	LLFF (4-view)			LLFF (6-view)			LLFF (9-view)		
		LPIPS*↓	PSNR↑	SSIM↑	LPIPS*↓	PSNR↑	SSIM↑	LPIPS*↓	PSNR↑	SSIM↑
NeRF-based	RegNeRF	29.65	18.55	0.587	22.61	19.08	0.760	18.39	22.86	0.820
	FreeNeRF	30.85	19.09	0.624	23.06	19.81	0.763	17.94	23.08	0.823
3DGS-based	3DGS	22.93	19.32	0.649	13.45	23.80	0.814	9.68	25.44	0.860
	CoR-GS	19.66	20.45	0.696	12.53	23.87	0.840	8.94	26.70	0.874
	DropGaussian	21.93	19.80	0.621	13.79	23.41	0.803	9.40	25.87	0.868
	D^{2}GS	17.98	22.35	0.746	13.85	23.91	0.846	8.91	26.80	0.871
	SynGS (Ours)	16.92	23.92	0.762	12.54	25.17	0.848	8.96	26.71	0.871

Mip-NeRF 360 and LLFF. Table 1 reports the results on Mip-NeRF 360 under the 4-, 6-, and 9-view settings. SynGS achieves the highest PSNR and SSIM in all three settings, with the largest PSNR gain in the 4-view case: 25.91 dB, 3.56 dB above D^{2}GS .[44] However, D^{2}GS obtains lower LPIPS values across the three settings, indicating an advantage in perceptual similarity. Table 2 shows a similar trend. With four views, SynGS leads in PSNR, SSIM, and LPIPS; under six views it obtains the highest PSNR and SSIM with LPIPS close to the best result. As the number of input views increases to nine, the differences among the leading methods become smaller: D^{2}GS [44] gives slightly higher PSNR and lower LPIPS, while CoR-GS [13] achieves slightly higher SSIM. Overall, SynGS shows more pronounced improvements in the four- and six-view settings, suggesting that the proposed framework is particularly beneficial when the geometric information available from the input views is severely limited.

Qualitative Results. Figure 3 shows the 4-view comparison on Mip-NeRF 360. SynGS produces fewer floating artifacts and more complete object boundaries, and preserves finer structures where baselines show broken contours or missing details.

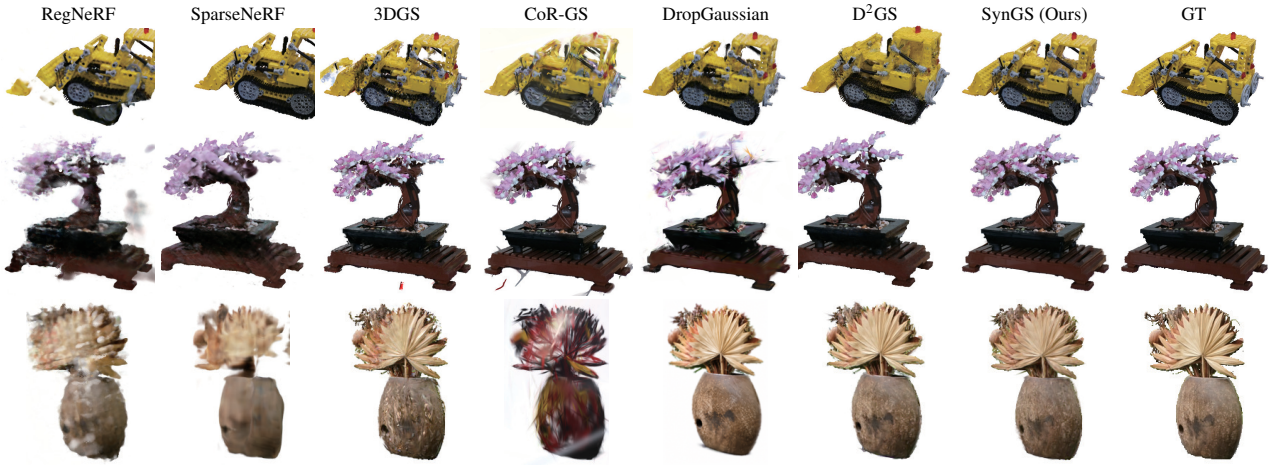


Figure 3. Qualitative comparison under the 4-view sparse-input setting on Mip-NeRF 360. SynGS produces fewer floating artifacts and more complete structural boundaries than the compared methods.

4.3 Ablation Study

Table 3 isolates the contributions of the three components. Unlike Table 1, which reports dataset-wide averages, the ablation is a separate protocol and is not the 4-view mean in Table 1.

Table 3. Ablation study of SynGS components. LPIPS* denotes LPIPS $\times 10^2$.

Visual Hull Init	Depth Densification	Visibility Balancing*	SSIM↑	PSNR↑	LPIPS*↓
√			0.865	17.51	12.80
√	√		0.902	24.96	5.20
√	√	√	**0.943**	**26.18**	**3.98**

*Visibility Balancing denotes Dynamic Visibility Regularization (DVR).

Visual Hull Initialization establishes the geometric baseline at 17.51 dB PSNR. Adding Depth Densification raises PSNR to 24.96 dB, the largest gain in the ablation. Adding DVR further improves the full configuration to 26.18 dB, with gains in SSIM and LPIPS.

5. CONCLUSION

We present SynGS for sparse-view 3DGS, combining **Visual Hull Initialization**, **VGGT-Guided Densification**, and **Dynamic Visibility Regularization** for more reliable initialization, geometry completion, and reduced sparse-view overfitting. SynGS demonstrates more pronounced improvements under highly sparse-view conditions, particularly in PSNR and SSIM with four or six input views. With nine input views, SynGS maintains improvements in PSNR and SSIM on Mip-NeRF 360, while achieving performance comparable to leading methods on LLFF. However, the additional geometry introduced during initialization incurs memory and computational overhead. Future work will focus on developing more efficient geometry augmentation strategies to reduce these costs while preserving reconstruction quality.

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